

Theory Course Code: BS102/BS202				Theory Course Name: Environmental Science and Waste Management								L	T	P
Course Offered in: CSE/IT/CS-AI/CS-AIML/CS-DS/EC												2	0	0
Course Objectives:														
The objective of this course is to provide B.Tech students with fundamental knowledge of environmental science and waste management, focusing on pollutants, sustainable technologies, waste classification, treatment and minimization strategies, and analytical techniques, enabling them to address engineering problems with an environmentally responsible and sustainable approach.														
Course Outcome: After completion of the course, the student will be able to														
1. CO1: To define and recall (Remember – L1) fundamental environmental science terms, environmental segments, and the nature of common atmospheric pollutants including smog, greenhouse gases, and radioactive materials. 2. CO2: To describe (Understand – L2) the principles of water and soil chemistry, including water treatment methods, nutrient cycles, and the use of nanomaterials in environmental cleanup. 3. CO3: To apply (Apply – L3) chemical principles to energy and fuel systems such as fuel cells, solar cells, and sustainable polymers, and to solve related numerical problems in fuel analysis. 4. CO4: To analyze (Analyze – L4) environmental risks due to fossil fuels, contaminated water, and solid/hazardous waste, and to evaluate appropriate waste management strategies.														
CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)														
CO-PO Mapping	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		
CO1	2	-	-	-	-	1	1	1	1	-	-	2		
CO2	1	1	1	-	-	1	2	-	-	-	-	1		
CO3	2	2	2	-	-	2	3	2	-	-	-	2		
CO4	-	2	1	-	-	2	2	-	-	-	-	2		
CO5	-	-	-	-	-	-	-	-	-	-	-	-		
Unit 1			Environmental Segments and Atmospheric Pollutants									08 hours		
Environmental segments: Composition and chemical characteristics of atmosphere, hydrosphere, lithosphere. Air pollutants: sox, nox, CO, vocs, Dioxins, furans. Smog formation: Sulphurous and photochemical smog (London smog) Ozone layer: Formation, depletion by cfcs/nox, and mitigation strategies. Greenhouse gases: CO ₂ , CH ₄ , N ₂ O – molecular properties and climate impacts.														
Unit 2			Natural Resources: Water and Soil management									08 hours		
Water quality parameters: pH, alkalinity, hardness and its determination, DO, BOD, COD. Water Technology: Techniques for water softening (Zeolite, Ion Exchange and Reverse Osmosis process), boiler troubles. Advance treatment of polluted water: Coagulation, flocculation, disinfection, Nanomaterials (e.g.,ZnO, TiO ₂ photocatalysts) in environmental cleanup. Soil quality parameters: Composition and properties, analysis of soil (N, P, K), bio fertilizers (Azotobacter, Rhizobium, Blue green algae.														
Unit 3			Conventional and Non-Conventional Energy Resources									08 hours		
Conventional Energy Resources: Fuel and combustion, Definition, Classification, Characteristics of a good fuel, Calorific Values, Gross & Net calorific value, Determination of calorific value by Bomb Calorimeter,														

Theoretical calculation of calorific value by Dulong's method, Ranking of Coal, Analysis of coal by Proximate and Ultimate analysis method

Non-Conventional Energy Resources: solar cells, Photovoltaics, biogas, biodiesel.

Unit 4	Environmental Toxicants and Solid Waste Remediation	08 hours
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Toxic chemicals: Types (mutagens, carcinogens, insecticides, pesticides)

Case studies: Industrial disasters (Bhopal Gas Tragedy, Shell oil spill, and Chernobyl nuclear disaster)

Solid and hazardous waste: Classification and management techniques

E-waste: leaching of heavy metals and their management.

Biodegradable polymer: PLA, PGA, PHBV, PVA, PCL

Unit 5	Environmental Monitoring and Management	08 hours
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Environmental Monitoring Techniques: Principles of chromatography and spectrophotometry techniques: paper chromatography, High Performance Liquid chromatography, UV- Visible spectroscopy.

Environmental management system (EMS): Environmental audit; Environmental clearance for establishing industries; Environmental Impact Assessment (EIA); EIA guidelines 1994.

Total Lecture Hours	40 hours
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Textbook:

1. Text book of Environment and Ecology by Dave, Katewa & Singh; Cengage India. (2022)
2. Environment and Ecology by S. Deswal & A. Deswal; Dhanpat Rai & Co. (2018)
3. Environment & Ecology by Smriti Srivastava; S.K. Kataria & Sons. (2017)
4. Engg. Chemistry Prof Sunita Rattan, S.K. Kataria & Sons. (2024)
5. Principles of Soil Chemistry by Kim. H. Tan (Taylor & Francis / CRC Press imprint). (2010)
6. Fundamental of Soil Science by Bijay Singh & Dilip Kumar Das; Indian Society of soil Science. (2012)
7. Textbook of Environment and Ecology by K. M. Gupta; Umesh Publication (2009)

Reference Books:

1. Environmental Studies: From Crisis to Cure by Rajagopalan; Oxford University Press. (2016)
2. Materials Chemistry by Fahlman B. D; Germany, Springer Netherlands, 2023.
3. Chemistry for Engineers by S.Vairam and Suba Ramesh; Wiley India, 2020
4. Environmental Chemistry by Stanley E. Manahan; CRC Press, 10th Ed.2017.

Justification of CO -PO Mapping

CO-PO	JUSTIFICATION
CO1 WITH PO1, PO6, PO7, PO8, PO9 and PO12	Learning fundamental environmental science concepts strengthens scientific and analytical foundations (PO1), promotes environmental consciousness (PO6, PO7). This enhances ability of an individual to contribute in team based activity for managing the project ethically (PO8 and PO9). Furthermore, It will foster the mindset required for lifelong learning in the context of evolving environmental challenges and technologies (PO12).
CO2 WITH PO1, PO2, PO3, PO6, PO7 and PO12	Understanding water and soil chemistry along with treatment technologies helps students to apply their theoretical knowledge (PO1 and PO2), in development of solutions for public health (PO3), describing sustainability strategies (PO6, PO7) and nurturing lifelong learning (PO12).
CO3 WITH PO1, PO2, PO3, PO6, PO7, PO8 and PO12	Application of knowledge in fuel cells, and green materials develops technical competency (PO1, PO2), supports basic design and solution development (PO3), encourages sustainable innovation (PO6, PO7), apply ethical principles (PO8) and promotes independent learning (PO12) in broadest perspective.
CO4 WITH PO2, PO3, PO6, PO7, and PO12	Analyzing chemical toxicants and interpreting data from instrumentation builds scientific understanding about designing (PO2, PO3) of enriching solutions. Use of research based knowledge to develop critical thinking and socio-economic responsibility (PO6, PO7) in managing the multidisciplinary projects suitable for the current and variable needs of environment (PO12).